Greenhouse gas from farm animals

Animals that chew cud, such as cattle and sheep, are called ruminants. A part of their intestines called the rumen, which is one of four stomachs, contains billions of bacteria and protozoa that help to break up grass into useful nutrients. This produces **methane gas**, which is a **greenhouse gas**, as a by-product. Greenhouse gas from farm animals accounts for about 10 per cent of Australia's greenhouse gas emissions. CSIRO scientists are trying to find ways to reduce emissions of methane gas from cattle and sheep. They have three projects underway:

- study of **gut microbes** in ruminants
- measuring methane emissions from cattle
- investigating how anti-microbial properties contained in different foods affect the microbes in the rumen.

☞ **1** Fill in the missing words to complete the sentences below.

Sheep and cattle produce _____ – _____ % of all of the greenhouse gas emissions in Australia. Some CSIRO scientists are studying g___ m_____ in ruminants. Other scientists are measuring _____ emissions from cattle in Northern Australia, while some investigate _____ properties of different foods.



Tammar Wallaby

2 Choose the correct word from each pair to complete the sentences below.

Native [grazers/predators] such as the Tammar Wallaby have **microbes** in their [pouch/foregut] that break down grasses differently and produce far less methane than ruminants. Scientists are [hopeful/frightened] that this knowledge will help them to [increase/reduce] methane production in cattle and sheep in future.

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